

Linear Algebra – Homework 4

Sections 3.1–3.3: Determinants, Cramer’s Rule, and Adjugate Matrices

Due: _____

Instructions: Show all necessary work. For determinant computations, clearly indicate the determinant properties, row operations, or cofactor expansions you use. When asked to justify an answer, a correct conclusion without supporting reasoning may receive limited credit.

1. Suppose A and B are 3×3 matrices with

$$\det(A) = -2, \quad \det(B) = 3.$$

Compute each of the following without finding the entries of A or B .

- (a) $\det(AB)$
- (b) $\det(A^T)$
- (c) $\det(A^{-1})$
- (d) $\det(4A)$
- (e) $\det(A^2B^{-1})$

2. Let A be a 3×3 matrix with $\det(A) = 5$. A matrix B is obtained from A by performing the following row operations, in order:

$$R_1 \leftrightarrow R_3, \quad R_2 \leftarrow -2R_2, \quad R_3 \leftarrow R_3 + 7R_1.$$

Find $\det(B)$. Explain the effect of each operation on the determinant.

3. Starting with a 3×3 matrix A , the following operations are performed:

$$R_1 \leftrightarrow R_3, \quad R_2 \leftarrow 3R_2, \quad R_3 \leftarrow R_3 - 2R_1,$$

producing

$$U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 4 \\ 0 & 0 & 2 \end{bmatrix}.$$

Find $\det(A)$ without reconstructing A .

4. Determine whether each statement below is **true or false**. Justify each answer. If a statement is false, give a counterexample or correct the statement.
- (a) $\det(A + B) = \det(A) + \det(B)$.
 - (b) If $\det(A) = 0$, then A must contain a zero row.
 - (c) If A and B are square matrices of the same size and $\det(AB) = 0$, then $\det(A) = 0$ or $\det(B) = 0$.
 - (d) If $\det(A) = \det(B)$, then $A = B$.
 - (e) Adding a multiple of one row to another row does not change the determinant.

5. Let

$$A_t = \begin{bmatrix} 1 & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{bmatrix}.$$

Use determinant properties and row operations, rather than immediately using a full cofactor expansion, to find $\det(A_t)$. For what value(s) of t is A_t invertible? Explain.

6. Compute

$$\det \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 0 & 1 & 1 & 2 \end{bmatrix}.$$

Use row replacement operations to reduce the matrix to upper triangular form. Do **not** begin with a 4×4 cofactor expansion.

7. Compute the determinant using a cofactor expansion. Choose the row or column that makes the computation most efficient, and state your choice.

$$\begin{vmatrix} 3 & 0 & 2 \\ 0 & -1 & 4 \\ 5 & 0 & 1 \end{vmatrix}.$$

8. Compute

$$\begin{vmatrix} 2 & 0 & 1 & 0 \\ 0 & 3 & 0 & 0 \\ 1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 5 \end{vmatrix}$$

using cofactor expansion. Your work should make clear why the locations of the zeros are useful.

9. Let

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & 5 \\ 2 & 1 & -1 \end{bmatrix}.$$

Find M_{12} , C_{12} , M_{23} , C_{23} , M_{31} , and C_{31} . Then compute $\det(A)$ by expanding along the second row.

10. Let

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 3 & -1 & 2 \\ 0 & 4 & 5 \end{bmatrix}.$$

- (a) Compute $\det(A)$ using the six-term formula for a 3×3 determinant.
- (b) Compute $\det(A)$ again using cofactor expansion along the second column.
- (c) Explain why the two methods must give the same result even though the intermediate calculations look different.

- 11.** Recall that in the general determinant formula, every product term contains exactly one entry from each row and exactly one entry from each column.
- (a) How many product terms are there in the determinant formula for a 4×4 matrix?
 - (b) Explain why $a_{11}a_{22}a_{33}a_{44}$ can occur as a term, while $a_{11}a_{22}a_{34}a_{44}$ cannot.
 - (c) Consider the product $a_{12}a_{24}a_{31}a_{43}$. The selected column order is $(2, 4, 1, 3)$. Determine whether this term appears with a $+$ sign or a $-$ sign by considering how many exchanges are needed to restore the order to $(1, 2, 3, 4)$.

- 12.** Let the rows of a 3×3 matrix A be r_1, r_2, r_3 , and suppose $\det(A) = 6$. A matrix B has rows

$$r_1 + 2r_2, \quad -3r_2, \quad r_3 - 5r_2.$$

Find $\det(B)$ without knowing any entries of A .

- 13.** Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix},$$

and let C_{ij} denote the cofactor of a_{ij} . Explain why

$$a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23} = 0.$$

Hint: Construct a matrix whose second row is replaced by the first row of A , and think about its determinant. Then explain why, in contrast,

$$a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} = \det(A).$$

- 14.** Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \\ 2 & 0 & 1 \end{bmatrix}.$$

Find

- (a) the cofactor matrix C ;
- (b) $\text{adj}(A)$;
- (c) $\det(A)$.

15. Using your results from Problem 14, find A^{-1} using

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Verify your result by checking one complete row-by-column product in AA^{-1} .

16. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Find $\operatorname{adj}(A)$, and verify directly by matrix multiplication that

$$A \operatorname{adj}(A) = \det(A)I_2.$$

Use this identity to explain why

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

requires $\det(A) \neq 0$.

17. A student argues:

If $\det(A) = 0$, then $A \operatorname{adj}(A) = 0$. Therefore $\operatorname{adj}(A) = 0$.

Explain why this reasoning is incorrect. Give a specific 2×2 matrix A such that $\det(A) = 0$ but $\operatorname{adj}(A) \neq 0$.

18. Suppose

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \operatorname{adj}(A) = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$$

- (a) Without expanding $\det(A)$ directly, use $A \operatorname{adj}(A) = \det(A)I$ to determine $\det(A)$.
 (b) Find A^{-1} .

19. Use **Cramer's Rule** to solve

$$\begin{cases} 2x - y = 1, \\ 3x + 2y = 8. \end{cases}$$

Clearly identify A , A_1 , and A_2 , and compute the necessary determinants.

20. Use **Cramer's Rule** to solve

$$\begin{cases} x + 2y - z = 6, \\ 2x + y + z = 3, \\ x - y + z = -2. \end{cases}$$

Show the determinants used to obtain x , y , and z .

21. Consider the system

$$\begin{cases} 2x_1 - x_2 + x_3 = 5, \\ x_1 + 2x_2 = -1, \\ 3x_1 + x_2 + 2x_3 = 6. \end{cases}$$

Use Cramer's Rule to find **only** x_2 . Do not solve for x_1 or x_3 .

22. Consider the system

$$\begin{cases} x + y + z = 1, \\ x + ty + z = 1, \\ x + y + tz = 1. \end{cases}$$

- (a) Use your result from Problem 5 to determine for which values of t Cramer's Rule can be used.
- (b) For those values of t , solve the system.
- (c) What happens when $t = 1$? Does the system have no solution, one solution, or infinitely many solutions? Explain.

23. Let A be an invertible $n \times n$ matrix and let $Ax = b$. Suppose A_j denotes the matrix obtained from A by replacing column j with b . Starting from

$$x = A^{-1}b = \frac{1}{\det(A)} \operatorname{adj}(A)b,$$

explain why the j -th component of $\operatorname{adj}(A)b$ is equal to $\det(A_j)$. Use this observation to explain why

$$x_j = \frac{\det(A_j)}{\det(A)}.$$

Your explanation should explicitly connect the matrix product with a cofactor expansion.

24. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

A student correctly computes the cofactor matrix as

$$C = \begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix},$$

but then writes $\operatorname{adj}(A) = C$.

- Identify the student's error.
- Find the correct $\operatorname{adj}(A)$.
- Compute both AC and $A \operatorname{adj}(A)$. Which product has the form $\det(A)I$?

25. Let A be a 3×3 real matrix satisfying

$$A^T = -A.$$

Prove that

$$\det(A) = 0$$

without expanding the determinant. Your proof may use $\det(A^T) = \det(A)$ and the effect of multiplying a 3×3 matrix by -1 .